

Next Generation Data Centers and Energy Management Systems

Lecture 4A: Applied Computational Fluid Dynamics (CFD)

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- **Applied Computational Fluid Dynamics (CFD)**
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Define the computational methods used for analyzing airflow in data center hall.
- Use finite difference and finite volume methods to analyze a canonical flow problem.

Typical layout of a data center facility

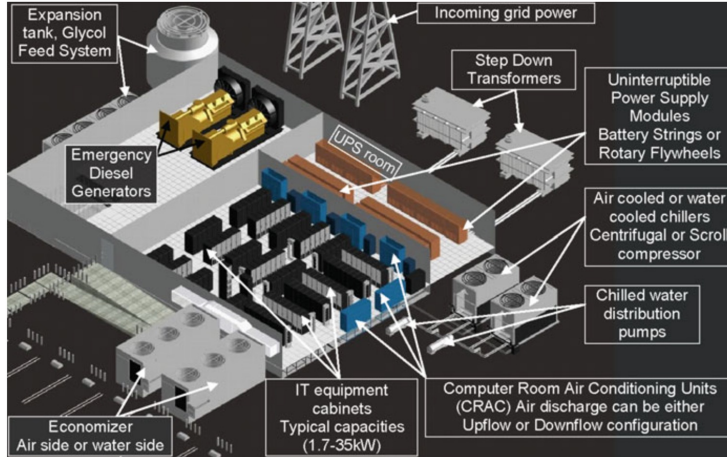


Figure 1: (Figure 1.1 in Ref. [2]).

Seminar 1

- Dr. Imran Latif. Johnson Controls International (JCI)
- Title: From Chip to Chiller: LLM Agents as the Next Efficiency Lever in AI Data Centers
- Date: Wednesday, 9/16/26
- Time: 9:00 am (Berkeley Time)
- Location: Etcheverry Hall 3110
- It will be recorded and posted to bCourses. Please do not distribute.

Project 2

In Project 2, you will analyze airflow within the data hall using CFD. You will develop airflow handling strategies. You will use Cadence Reality DC Design Pro.

Computational Methods in CFD

Derivation comes from Ref.[\[1\]](#).

Finite Difference: Lid Driven Cavity

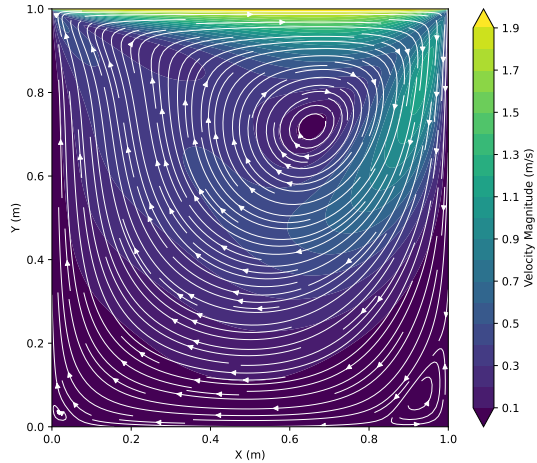


Figure 2: Compare with Figure 7.8 from Ref.[1].

Finite Volume: Lid Driven Cavity

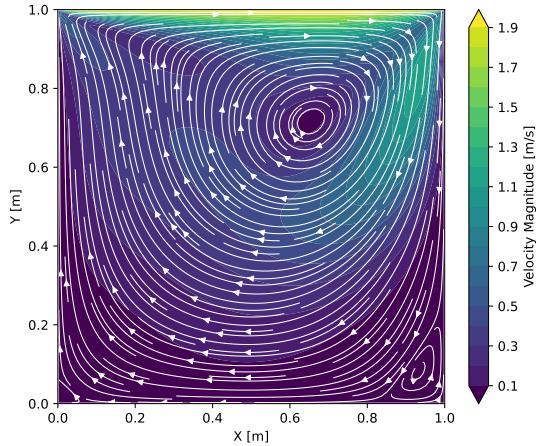


Figure 3: Compare with Figure 7.8 from Ref.[1].

Finite Difference: Buoyance-driven Cavity

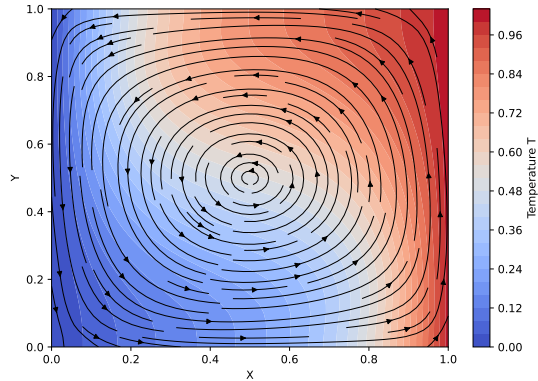


Figure 4: Compare with Figures 7.8 and 7.17 from Ref.[1].

References I

- [1] J. H. Ferziger and M. Perić.
Computational Methods for Fluid Dynamics.
Springer, 3 edition, 2002.
- [2] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.